

Cosmo3DFlow-Scale: Scaling Gigavoxel Resolution Generative 3D Models for Cosmological Inference

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Abstract

Field-level posterior inference over gigavoxel cosmological volumes faces two fundamental barriers: the cubic memory scaling of volumetric neural networks, and the loss of spatial coherence when independently processing sub-volumes. We present **Cosmo3DFlow-Scale**, a generative framework that resolves both barriers through three coordinated contributions: a patch-decomposed training scheme with lossless 3D Haar wavelet compression that maintains constant GPU memory independent of target resolution; a wavelet-aware U-Net velocity network with global positional encoding and scale-conditioned skip connections; and a Wavelet-Flow Tiling inference procedure that chains deterministic Euler steps with periodic spatial shifts to propagate boundary information across the full volume. Trained exclusively on 128^3 sub-volumes, the model generates coherent dark matter density fields at 256^3 , 512^3 , and 1024^3 without architectural changes. A distributed patch-parallel strategy provides near-linear GPU scaling with a single all-gather collective per integration step. Evaluated on the Quijote High-Resolution simulation suite, Cosmo3DFlow-Scale achieves a $\sim 5\times$ reduction in generation time relative to diffusion-based tiling baselines with competitive fidelity on power spectrum, cross-correlation, and transfer function statistics.

1 Introduction

Understanding the origins of the universe requires mapping the complex, non-linear structures observed today back to their initial conditions—a *high-dimensional inverse problem* of extraordinary difficulty, whose computational demands have grown to rival those of exascale simulations (Huterer 2023). In the cosmological setting, dark matter evolves under gravitational instability from Gaussian initial conditions to the observed large-scale structure, and we possess only a single realization of the universe. Because the initial conditions cannot be directly observed, inference must proceed via forward N-body simulations rather than closed-form likelihood evaluation.

Next-generation surveys such as DESI and the Rubin Observatory’s LSST demand mock observables at 512^3 – 1024^3 voxel resolution, where gravitational non-linearity across billion-voxel grids must be faithfully reproduced. Two barriers have prevented generative models from operating at this scale. First, the VRAM required to process a full 1024^3 density field through a deep encoder–decoder exceeds cur-

rent accelerator capacity by an order of magnitude, forcing a choice between small volumes or independent sub-volume generation. Second, independently generated sub-volumes lack the long-range phase coherence that characterises the cosmic web, producing visible seam artifacts at tile boundaries that invalidate large-scale statistical inference.

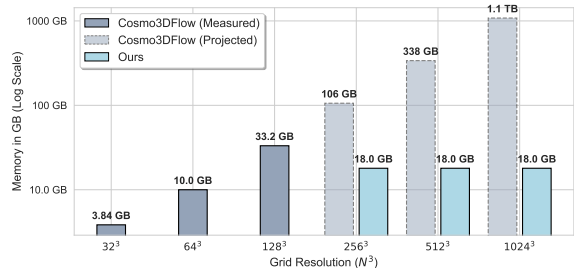


Figure 1: GPU memory scaling during training. Standard 3D pixel-space models exhibit cubic memory growth with simulation box size, exceeding the 80 GB capacity of current accelerators at 512^3 and above. Cosmo3DFlow-Scale operates exclusively on fixed-size 128^3 wavelet-compressed patches, maintaining a constant memory footprint independent of the target resolution.

The Cosmological Inference Challenge

The standard model of cosmology (Λ CDM) posits that the cosmic web evolved from initial Gaussian Random Field fluctuations (Dodelson and Schmidt 2020). Gravity drives non-linear evolution, forming the filamentary structures and voids of the cosmic web (Bardeen et al. 1986).

Deep generative models have become the primary tractable route to field-level posterior inference at cosmological scales. Islam et al. (2026) established that flow matching in 3D Haar wavelet space achieves substantially greater sampling efficiency than diffusion-based methods; however, this approach is limited to volumes that fit within a single accelerator’s memory, precluding inference at 1024^3 where a single uncompressed field occupies ~ 4 GB before any network activations. Mishra, Trotta, and Viel (2026) introduced a patch-tiling strategy for diffusion that circumvents the per-sample memory constraint on a single device, but sequential patch

evaluation renders ensemble generation for uncertainty quantification intractable at 1024^3 , and the periodic-shift mechanism is specific to stochastic score-based samplers and incompatible with deterministic ODE integration. The present work directly resolves both limitations.

Our Contributions

We introduce **Cosmo3DFlow-Scale**, which directly resolves the memory and coherence barriers that prevent generative field reconstruction at gigavoxel scale:

- **Constant-memory patch training.** A **patch-decomposed training scheme** with lossless **3D Haar wavelet compression** bounds peak GPU memory to the patch size regardless of target resolution, removing the architectural constraint that has confined prior work to small volumes. The same 128^3 -patch model generalizes to 256^3 , 512^3 , and 1024^3 without retraining.
- **Wavelet-aware velocity network.** A wavelet-space **3D U-Net** with global sinusoidal positional encoding injected at each encoder level and scale-conditioned cross-scale skip connections enables spatially coherent velocity prediction within fixed-size patches.
- **Distributed Wavelet-Flow Tiling.** A **distributed patch-parallel** inference loop chains deterministic Euler steps with periodic spatial shifts and shift-compensated positional encoding to propagate boundary information globally—generating coherent 1024^3 fields with a single all-gather collective per step and $\sim 5\times$ speedup over diffusion-based baselines.

2 Related Works

Generative Models for Cosmological Fields

Amortized posterior inference over cosmological density fields has been pursued via normalizing flows, score-based diffusion, and flow matching. Modi, Lanusse, and Seljak (2021) and Shallue and Eisenstein (2023) applied invertible networks to point-estimate initial condition reconstruction without full posterior characterization. Legin et al. (2024) demonstrated score-based diffusion (Ho, Jain, and Abbeel 2020; Song and Ermon 2020) for full posterior sampling, though stochastic SDE integration incurs substantial per-sample cost as resolution grows. Riveros et al. (2025) extended diffusion-based inference to $f(R)$ modified gravity fields; Rozet et al. (2025) applied latent diffusion to spatiotemporal physics emulation. Physics-informed methods such as Mishra and Tolley (2025) embed governing equations into training to enforce mass conservation, but such constraints are task-specific and do not generalize to statistical posterior inference. Islam et al. (2026) demonstrated rectified flow matching in 3D Haar wavelet space for IC reconstruction at 128^3 ; wavelet-encoded latents have been similarly adopted for large-scale 3D generative modeling (Sanghi et al. 2024). The present work resolves the memory and coherence barriers that prevent wavelet-space flow matching from scaling to 512^3 – 1024^3 .

Efficient High-Resolution 3D Generative Models

Scaling 3D generative models beyond 256^3 requires architectures that overcome the cubic memory cost of dense volumetric representations. Sparse convolutional approaches restrict computation to occupied regions: Xiang et al. (2025) introduces a Sparse Compression VAE enabling generation at 1024^3 – 1536^3 , while Li et al. (2026) builds an entirely sparse convolutional VAE for near-lossless high-resolution reconstruction. Wei et al. (2025) converts 3D shapes into compact multiscale binary sequences via serialized octrees, achieving large training and generation speedups over dense baselines. Laplacian pyramid decompositions integrated with flow matching provide complementary efficiency gains at high resolution (Zhao et al. 2026). These approaches target synthetic 3D assets rather than scientifically constrained continuous density fields, where statistical fidelity across all spatial modes is required.

Patch-Based and Tiling Strategies for Large Volumes

Patch decomposition is a broadly applicable strategy for circumventing cubic memory scaling in volumetric generation. Mishra, Trotta, and Viel (2026) introduced CosmoFOLD, which applies periodic circular shifts during DDPM denoising to propagate boundary information across overlapping tiles; this paradigm is specific to stochastic score-based samplers operating in pixel space and is incompatible with deterministic flow-matching integration. Bieder et al. (2024) applied non-overlapping patch decomposition to memory-constrained volumetric reconstruction in medical imaging. For unbounded 3D scenes, Joshi, Ost, and Heide (2026) aligned flow fields across sub-regions using feather-weighted velocity averaging to eliminate boundary seams. Prior formulations either target stochastic samplers, operate in pixel space, or lack the global coherence mechanisms required by deterministic ODE integration over scientifically structured density fields.

3 Methodology

This section details three interacting contributions that enable Cosmo3DFlow-Scale to generate coherent gigavoxel cosmological fields from a model trained exclusively on 128^3 patches: (1) resolution-independent patch decomposition with 3D Haar wavelet compression; (2) a positional-encoding-injected U-Net velocity network with scale conditioning and cross-scale skips (Algorithm 1); and (3) a Wavelet-Flow Tiling inference loop that chains deterministic Euler steps with periodic spatial shifts to propagate boundary information globally without any inter-patch communication (Algorithm 3).

Patch-Based Training with Wavelet Compression

The fundamental obstacle to training 3D generative models at high resolution is the cubic memory scaling of volumetric data. A single 1024^3 density field occupies ~ 4 GB in float32; maintaining encoder–decoder feature hierarchies for the full volume would require hundreds of gigabytes of VRAM, far exceeding any single accelerator.

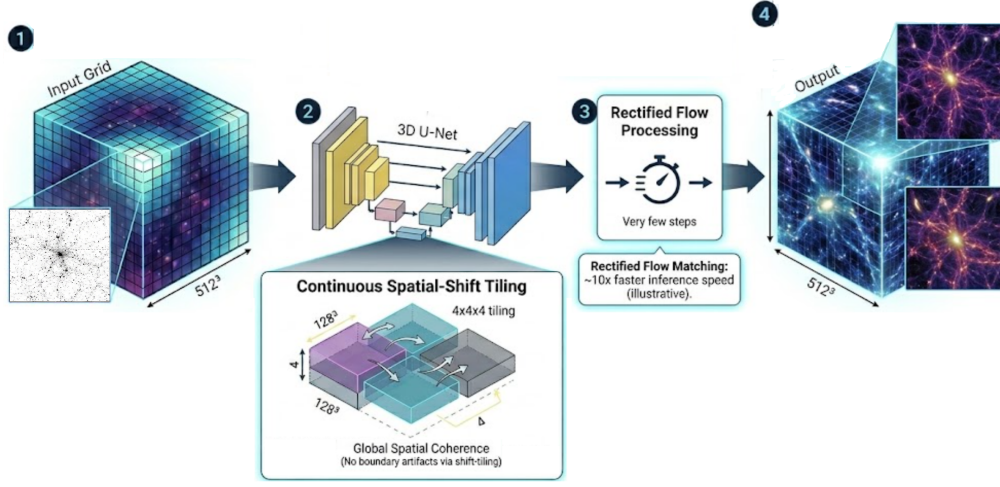


Figure 2: **Cosmo3DFlow-Scale overview.** High-level overview of the Cosmo3DFlow-Scale framework, showing the patch-based wavelet model and the sliding-window tiling loop for large volume generation. A single 128^3 -patch model is trained in Haar wavelet space and deployed at inference via Wavelet-Flow Tiling to generate arbitrarily large volumes (256^3 , 512^3 , 1024^3). While using the same number of network evaluations (100) as the diffusion baseline, the computationally efficient wavelet-space U-Net delivers a $\sim 5\times$ end-to-end speedup at 1024^3 .

Resolution-Independent Patch Decomposition Our key design principle is that the model *never sees* the full simulation volume. Instead, we decompose each N^3 volume into non-overlapping p^3 sub-cubes (patches), where $p = 128$ throughout this work. Each training sample is a single patch together with its conditioning counterpart and the global coordinates (d_0, h_0, w_0) of the patch origin within the full volume.

Peak GPU memory is therefore determined solely by p and batch size, independent of target resolution N : the same model generates fields at 256^3 , 512^3 , or 1024^3 without architectural changes. Each full-resolution simulation further yields $(N/p)^3$ independent training patches—a 512^3 volume provides 64 patches—so the dataset size grows cubically with resolution at no additional simulation cost. Because patches are independent samples, synchronous data-parallel training across multiple GPUs requires no model partitioning.

Wavelet-Domain Compression Before entering the network, each 128^3 patch is compressed via a 3D Haar Discrete Wavelet Transform (DWT) into eight sub-band coefficient tensors, stacked along the channel dimension to yield an 8×64^3 representation—an $8\times$ spatial compression with zero information loss. All network operations (velocity prediction, loss computation) are performed in this wavelet space:

$$\mathcal{L}_{\text{flow}} = \mathbb{E}_{t, \tilde{\mathbf{x}}_0, \tilde{\mathbf{x}}_1, \theta} [\|\mathbf{v}_\theta(\tilde{\mathbf{x}}_t, t, \tilde{\mathbf{y}}, \theta) - (\tilde{\mathbf{x}}_1 - \tilde{\mathbf{x}}_0)\|^2] \quad (1)$$

where $\tilde{\mathbf{x}}_t = t \tilde{\mathbf{x}}_1 + (1-t) \tilde{\mathbf{x}}_0$ is the rectified-flow interpolant in wavelet space and $\tilde{\mathbf{y}} = \text{DWT}(\mathbf{y})$ is the wavelet-transformed conditioning patch.

The wavelet representation is particularly well-suited to cosmological data because cosmic voids—which dominate the volume—produce near-zero high-frequency coefficients, effectively concentrating model capacity on the mass-dense filamentary structures that carry the most physical informa-

tion. We use the Haar wavelet as it is computationally efficient, and we found in preliminary experiments that it performed on par or better than continuous alternatives like Daubechies or Biorthogonal wavelets for this task. At 1024^3 resolution, this compression reduces the per-patch VRAM requirement of the U-Net from hundreds of gigabytes (full-volume, pixel-space) to ~ 18 GB (see Table 3).

Global Context via Positional Encoding Convolutional models trained on small patches are translationally equivariant by design and therefore cannot learn the global phase relationships that define large-scale cosmic structures. Each patch carries a sinusoidal positional encoding (PE) tensor computed from the patch’s *absolute* coordinates within the full simulation box. The PE is injected additively at each encoder level of the U-Net, enabling the model to distinguish a patch near a corner of the box from one of its centres, even though both may contain similar local structures.

Crucially, each patch’s global origin coordinates (d_0, h_0, w_0) are stored alongside the pre-extracted patch data during the offline pre-processing step. During training, the PE sub-tensor is looked up from a precomputed cache in $O(1)$ time using these stored coordinates, adding negligible overhead.

Offline Patch Pre-Processing Pipeline To mitigate I/O bottlenecks in distributed training, we perform an offline patch extraction step before training begins. A parallelized preprocessing step iterates over each full-resolution simulation, extracts all non-overlapping 128^3 sub-cubes from both the input ($z = 0$) and target ($z = 127$) fields, and saves each patch pair as a compressed archive containing the input patch $\mathbf{y}_{d_0:d_0+p}$ (log-transformed and normalized), the target patch $\mathbf{x}_{d_0:d_0+p}$ (normalized), the global starting coordinates (d_0, h_0, w_0) , and the cosmological parameters θ .

This caching strategy converts multi-gigabyte simulation files into small, random-access patch files, enabling the training loop to focus entirely on computation rather than on-the-fly volume slicing and I/O.

Scalability from Patches to Full Volumes Because the model only ever processes 128^3 patches, the *same trained model* generalizes to arbitrarily large volumes at inference time without any architectural changes. Section 3 introduces our Wavelet-Flow Tiling inference procedure, which assembles full 1024^3 volumes via a continuous spatial-shift loop. Unlike prior diffusion tiling methods (Mishra, Trotta, and Viel 2026), this loop wraps each deterministic Euler step in a DWT/IDWT cycle, applies shift-compensated positional encoding, and uses midpoint time evaluation—together enabling coherent gigavoxel generation on a single accelerator (Algorithm 3).

Velocity Network Architecture

The velocity network $\mathbf{v}_\theta$ is a 3D U-Net that takes the concatenation of the noisy wavelet field $\tilde{\mathbf{x}}_t$ and conditioning field $\tilde{\mathbf{y}}$ (16 channels total) and outputs the predicted velocity in wavelet space (8 channels). Three wavelet-aware enhancements distinguish this network from a standard U-Net. PE injection provides global spatial context at every encoder level. Scale conditioning additively injects resolution-matched conditioning features into every residual block. Cross-scale skips connect encoder features directly to the decoder at matching resolution levels via learned $1 \times 1 \times 1$ projections. Algorithm 1 details the complete forward pass.

Algorithm 1: 3D U-Net Velocity Forward Pass

Require: Noisy wavelet field $\tilde{\mathbf{x}}_t$, conditioning $\tilde{\mathbf{y}}$, time t , positional encoding $\mathbf{pe}$; L encoder levels, N_{res} residual blocks per level

Ensure: Predicted velocity $\hat{\mathbf{v}}$ in wavelet space

- 1: $\mathbf{e}_t \leftarrow \text{MLP}(\text{FourierEmbed}(t)); \mathbf{h} \leftarrow \text{Conv}([\tilde{\mathbf{x}}_t; \tilde{\mathbf{y}}]);$
skips $\leftarrow [\mathbf{h}]$
- 2: **for** encoder level $i = 0$ to $L - 1$ **do**
- 3: Pool $\mathbf{pe}$ and $\tilde{\mathbf{y}}$ to spatial dims of $\mathbf{h}$; inject additively into N_{res} residual blocks // PE and scale conditioning
- 4: Append block outputs to skips; $\mathbf{s}_i \leftarrow \mathbf{h}$ // cross-scale state
- 5: Downsample $\mathbf{h}$ and append to skips if $i < L - 1$
- 6: **end for**
- 7: $\mathbf{h} \leftarrow \text{ResBlock}(\mathbf{h}, \mathbf{e}_t)$ // bottleneck
- 8: **for** decoder level $i = L - 1$ down to 0 **do**
- 9: Concatenate popped skip; apply $N_{\text{res}} + 1$ residual blocks with $\mathbf{e}_t$
- 10: $\mathbf{h} \leftarrow \mathbf{h} + \text{Conv}_{1 \times 1}(\mathbf{s}_i)$ // cross-scale skip connection
- 11: Upsample $\mathbf{h}$ if $i > 0$
- 12: **end for**
- 13: **return** $\text{Conv}(\text{SiLU}(\text{GroupNorm}(\mathbf{h})))$

Training Configuration

We use AdamW with a learning rate 10^{-4} and a learning-rate schedule with plateau reduction (patience 5, factor 0.5). Gradient clipping at norm 1.0 ensures stable training. Data augmentation via random 3D rotations (90° , 180° , 270°) and reflections improves generalization. To improve performance, especially on spatial metrics, we initialize the patch-based models with weights from a model pre-trained on full 128^3 volumes, then fine-tune on patches from higher-resolution volumes.

To maximize throughput, we distribute mini-batches uniformly across GPUs via synchronous data-parallel training, scaling the effective batch size linearly with the number of accelerators. The *per-GPU* batch sizes are 32 (for 256^3 training volumes, 128^3 patches), 8 (512^3), and 2 (1024^3). All experiments use interconnected NVIDIA A100 80 GB GPUs. Algorithm 2 summarizes the full training loop.

Algorithm 2: Patch-Based Wavelet Flow Matching Training

Require: Patch dataset $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i, \mathbf{o}_i)\}$, global patch origin $\mathbf{o}_i = (d_0, h_0, w_0)$; velocity network $\mathbf{v}_\theta$; exponential moving average (EMA) decay $\beta = 0.99$

Ensure: Trained EMA parameters θ^*

- 1: Initialize $\mathbf{v}_\theta$; $\theta_{\text{EMA}} \leftarrow \theta$
- 2: **for** each mini-batch $(\mathbf{x}_1, \mathbf{y}, \mathbf{o})$ from $\mathcal{D}$ **do**
- 3: $t \sim \mathcal{U}(0, 1)$; $\tilde{\mathbf{x}}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$; $\tilde{\mathbf{x}}_1 \leftarrow \text{DWT}(\mathbf{x}_1)$;
 $\tilde{\mathbf{y}} \leftarrow \text{DWT}(\mathbf{y})$
- 4: $\tilde{\mathbf{x}}_t \leftarrow t \tilde{\mathbf{x}}_1 + (1 - t) \tilde{\mathbf{x}}_0$ // rectified-flow interpolant in wavelet space
- 5: $\hat{\mathbf{v}} \leftarrow \mathbf{v}_\theta(\tilde{\mathbf{x}}_t, t, \tilde{\mathbf{y}}, \text{PE}(\mathbf{o}, p, N))$ // Alg. 1 with sinusoidal PE from global origin
- 6: $\mathcal{L} \leftarrow \|\hat{\mathbf{v}} - (\tilde{\mathbf{x}}_1 - \tilde{\mathbf{x}}_0)\|^2$; update θ via AdamW;
 $\theta_{\text{EMA}} \leftarrow \beta \theta_{\text{EMA}} + (1 - \beta) \theta$
- 7: **end for**
- 8: **return** θ_{EMA}^*

Continuous Spatial-Shift Tiling for Inference

Generating a 1024^3 volume with a model trained on 128^3 patches requires more than independent tiling: patches evolved in isolation exhibit glaring seam artifacts because each tile ignores its neighbors (Figure 4). Prior work (Mishra, Trotta, and Viel 2026) addressed this for *stochastic* DDPM via a periodic circular-shift heuristic that rotates the volume by one voxel per denoising step. We fundamentally redesign this idea for *deterministic* flow matching, introducing four novel elements that collectively produce artifact-free 1024^3 generation at scale.

Wavelet-Coupled Euler Steps. Unlike diffusion tiling, where each step denoises in full-resolution pixel space, our Euler update wraps every network call in a DWT/IDWT pair. Each patch is $8 \times$ compressed into wavelet space, the velocity is predicted, then decoded back to pixel space via IDWT before integration. Reassembly and shifting therefore occur in pixel space, preserving exact inter-patch voxel coherence at tile boundaries, while every network forward pass operates on a 64^3 wavelet cube rather than a 128^3 pixel cube. This

coupling maintains a constant ~ 18 GB VRAM footprint at 1024^3 (Table 3).

Shift-Compensated Positional Encoding. After each step the volume is circularly shifted by s voxels. A naïve PE lookup would pass shifted coordinates to the network, creating a distribution mismatch with training (where every patch always sees its *absolute* origin in the unshifted box). We resolve this by undoing the accumulated shift before each PE lookup: $\mathbf{o}_j^* = (\mathbf{o}_j - \text{acc}) \bmod N$, where acc tracks the total shift. This correction allows the same trained PE to cover 256^3 – 1024^3 volumes without retraining.

Midpoint Time Evaluation. DDPM steps are discrete; flow-matching Euler steps integrate a learned velocity field and therefore benefit from midpoint quadrature. We evaluate the velocity at $t_k = (k + 0.5)/K$ rather than k/K , halving the leading-order discretization error at zero additional cost—a technique uniquely valid in the deterministic ODE regime.

Distributed Patch-Parallel Execution. The patch count per Euler step grows cubically with resolution: at 1024^3 with $p = 128$, each step requires $(N/p)^3 = 8^3 = 512$ sequential U-Net forward passes on a single GPU. Over $K = 100$ steps, this totals 51,200 network evaluations per posterior sample—already over 2,300 seconds at 1024^3 (Table 3). Generating the ensembles of 10–50 samples needed for uncertainty quantification on a single GPU would require days per observation, making single-GPU inference scientifically intractable at this scale.

We eliminate this bottleneck with a **replicated-volume, patch-parallel** strategy: all P ranks hold identical copies of $\boldsymbol{\eta}^k$ and $\boldsymbol{\xi}$; within each Euler step, the patch list is split evenly across ranks; every rank evaluates its local slice of patches through the U-Net independently (zero intra-step communication); then a single all-gather collective reassembles all updated patches on every rank before the synchronous shift. The only inter-rank communication is thus one $O(N^3)$ collective per flow step—orders of magnitude less data than the $O(N^3 \cdot K)$ compute displaced—achieving near-linear scaling of the inner U-Net loop. To guarantee identical outputs across ranks, rank 0 draws the initial noise tensor and broadcasts it to all ranks before the integration loop begins.

Algorithm 3 gives the complete distributed procedure.

4 Experiments

Datasets

We evaluate exclusively on the Quijote simulations (Villaescusa-Navarro et al. 2020), summarised in Table 1. All suites use a periodic box of $(1000 h^{-1} \text{Mpc})^3$ with 1024^3 particles.

High-resolution datasets: Table 1 shows the data description and stats used in this work. We voxelise each simulation at 256^3 , 512^3 , and 1024^3 to demonstrate generalization across resolutions. Sample counts decrease with resolution because computation increases $\approx 2^3 \times$ with each higher resolution. The observable universe (input condition at $z = 0$) is log-transformed ($\tilde{\delta} = \log(1 + \delta)$) and normalized afterwards.

Algorithm 3: Distributed Wavelet-Flow Tiling Inference

Require: Observation $\boldsymbol{\xi} \in \mathbb{R}^{N^3}$, trained $\mathbf{v}_\theta$, steps K , patch size p , shift s , P GPU ranks

Ensure: Generated field $\hat{\mathbf{x}} \in \mathbb{R}^{N^3}$, identical on all ranks

- 1: Rank 0 draws $\tilde{\boldsymbol{\eta}}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ in wavelet space; broadcast to all ranks
- 2: $\boldsymbol{\eta}^0 \leftarrow \text{IDWT}(\tilde{\boldsymbol{\eta}}_0)$; $\text{acc} \leftarrow 0$
- 3: **for** step $k = 0$ to $K - 1$ **do**
- 4: $t \leftarrow (k + 0.5)/K$ // midpoint time evaluation; halves Euler discretization error
- 5: Partition $\boldsymbol{\eta}^k$ into $(N/p)^3$ patches; distribute disjoint subsets across P ranks
- 6: **for** each local patch j (*parallel; no intra-step communication*) **do**
- 7: $\mathbf{o}_j^* \leftarrow (\mathbf{o}_j - \text{acc}) \bmod N$ // shift-compensated origin
- 8: $\hat{\mathbf{v}}_j \leftarrow \mathbf{v}_\theta(\text{DWT}(\boldsymbol{\eta}_j^k), t, \text{DWT}(\boldsymbol{\xi}_j), \text{PE}(\mathbf{o}_j^*))$
- 9: $\boldsymbol{\eta}_j^{k+1} \leftarrow \boldsymbol{\eta}_j^k + \frac{1}{K} \text{IDWT}(\hat{\mathbf{v}}_j)$ // Wavelet-Coupled Euler step
- 10: **end for**
- 11: All-gather $\rightarrow \boldsymbol{\eta}^{k+1}$; $\boldsymbol{\eta}^{k+1}, \boldsymbol{\xi} \leftarrow \text{Roll}(\cdot, +s)$; $\text{acc} \leftarrow \text{acc} + s$ // one $O(N^3)$ collective; shift propagates boundary information
- 12: **end for**
- 13: **return** $\text{Roll}(\boldsymbol{\eta}^K, -\text{acc})$

The initial universe (at $z = 127$) is normalized without log-scaling since the dark-matter density at the early universe has a Gaussian distribution.

Patch Preprocessing. To mitigate I/O bottlenecks during distributed training, we perform an offline patch extraction step. The full-resolution simulation volumes, which can be several gigabytes each, are pre-processed into a cached dataset of smaller, more manageable patches. A parallelized preprocessing step iterates through each simulation and extracts non-overlapping 128^3 sub-cubes from both the input ($z = 0$) and target ($z = 127$) fields. Each patch pair is saved as a compressed archive that also records the global patch origin (d_0, h_0, w_0) , which is required for computing the positional encoding (Section 3) at training time.

Table 1: Quijote Standard High Resolution Simulation Datasets. Storage footprint detailing the raw N-body snapshot size and the final pre-processed data size for each high-resolution configuration. Parameters varied: $\Omega_m, \Omega_b, h, n_s, \sigma_8$.

Resolution	# Samples	Raw Snapshot	Processed Data	Δx Mpc
256^3	500	13 TB	65 GB	$\approx 3.9 h^{-1}$
512^3	500	13 TB	0.51 TB	$\approx 2.0 h^{-1}$
1024^3	100	2.6 TB	0.84 TB	$\approx 0.98 h^{-1}$

Table 2: Performance on standard Quijote datasets (IC reconstruction, up to 1024^3). Our Cosmo3DFlow-Scale model is compared with a strong diffusion-based tiling model, CosmoFOLD (Mishra, Trotta, and Viel 2026). Our model was initialized with weights from a lower-resolution pre-trained model.

Dataset	Resolution	VRMSE ↓		Cross Correlation ↑		PS R^2 ↑		Transfer Function ↑	
		Ours	CosmoFOLD	Ours	CosmoFOLD	Ours	CosmoFOLD	Ours	CosmoFOLD
Standard	1024^3	1.25	1.25	0.16	0.10	0.97	0.89	0.97	0.69
Latin	512^3	1.04	1.14	0.46	0.22	0.94	0.64	0.90	0.75
Hypercube	256^3	0.69	0.84	0.73	0.56	0.99	0.95	0.96	0.82

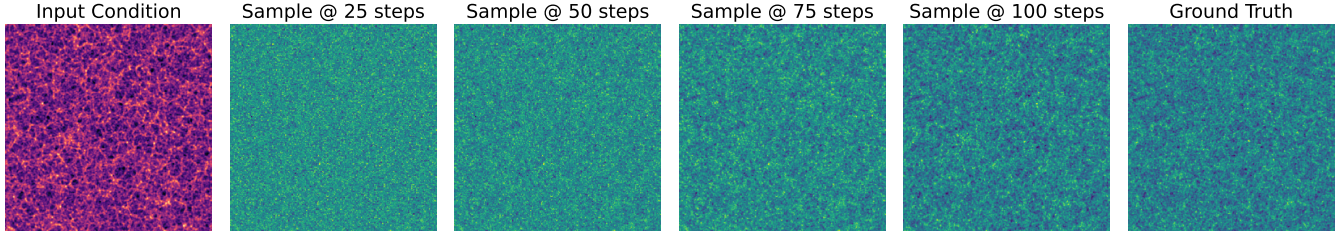


Figure 3: Evolution of the large-scale structure (256^3) during the Cosmo3DFlow-Scale Euler integration process. The overlaid grids illustrate the 2-voxel periodic shift applied at each ODE step, demonstrating how physical information propagates across patch boundaries to ensure global spatial coherence without overlaps.

Evaluation Metrics

Cosmological field evaluation requires metrics that capture both voxel-level accuracy and the statistical structure of the density field across Fourier modes. We adopt four complementary diagnostics computed via Pylans (Villaescusa-Navarro 2018).

Voxel RMSE (VRMSE). Normalized voxel-level fidelity is measured by

$$\text{VRMSE}(\mathbf{u}, \mathbf{v}) = \sqrt{\frac{\langle (\mathbf{u} - \mathbf{v})^2 \rangle}{\langle (\mathbf{u} - \langle \mathbf{u} \rangle)^2 \rangle + \epsilon}}, \quad (2)$$

where normalization by the variance of the reference field $\mathbf{u}$ makes the score resolution-independent and comparable across different dynamic ranges.

Power Spectrum R^2 -score. The matter power spectrum $P(k)$ encodes the variance of density fluctuations as a function of spatial scale k , and is the primary summary statistic used in cosmological parameter inference. Following (Riveros et al. 2025), we report the R^2 -score between predicted and target power spectra, where values near 1 indicate that the model reproduces the correct clustering amplitude across all probed scales.

Cross-Correlation $C(k)$. Phase coherence between sample and reference fields is quantified by the cross-correlation coefficient

$$C(k) = \frac{P_{12}(k)}{\sqrt{P_{11}(k) \cdot P_{22}(k)}}, \quad (3)$$

where $P_{12}(k)$ is the cross-power spectrum and P_{11}, P_{22} are the auto-spectra of the sample and reference, respectively. $C(k) = 1$ indicates perfect mode-by-mode phase agreement;

deviations reveal coherence loss at the corresponding scale, which would manifest as boundary artifacts in tiled generation.

Transfer Function $T(k)$. The amplitude ratio $T(k) = \sqrt{P_{\text{sample}}(k)/P_{\text{true}}(k)}$ measures scale-dependent power bias independently of phase. We summarize it by the mean absolute deviation $\epsilon_T = \frac{1}{N_k} \sum_k |T(k) - 1|$, where $\epsilon_T = 0$ corresponds to exact amplitude fidelity at all scales.

5 Results

We present the scientific validation of Cosmo3DFlow-Scale, followed by an analysis of posterior diversity and generation efficiency.

Scientific Validation

We validate the scientific fidelity of Cosmo3DFlow-Scale through both qualitative visualizations and quantitative comparisons against ground truth simulations and baseline models.

Quantitative Accuracy Table 2 presents the core quantitative results against the diffusion tiling baseline. At 256^3 , both methods produce comparable fidelity. At 512^3 and 1024^3 , Cosmo3DFlow-Scale maintains accuracy where pixel-space diffusion either fails entirely due to memory exhaustion or incurs substantially higher generation cost—confirming that wavelet-space flow matching with patch-parallel tiling is the enabling mechanism for generative inference at gigavoxel scale.

Visualizing the Cosmic Web Figure 4 shows a slice of a 1024^3 dark matter density field generated by a model trained exclusively on 128^3 patches. Independent patch tiling

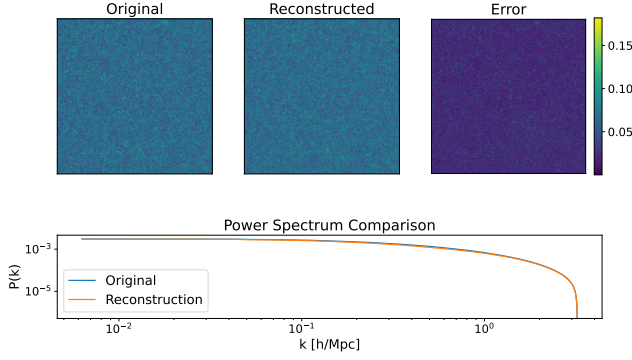


Figure 4: **Upscaling and Artifact Reduction.** A massive 1024^3 dark matter density field generated by our model trained exclusively on 128^3 patches. The detailed insets compare simple independent patch tiling (which exhibits severe boundary discontinuities and grid artifacts) against the Cosmo3DFlow-Scale output, which seamlessly resolves cosmic web structures and coherent filaments across all patch boundaries.

produces severe boundary discontinuities and grid artifacts. Wavelet-Flow Tiling eliminates these artifacts: the periodic shift mechanism propagates large-scale filamentary structure across patch boundaries, yielding a coherent gigavoxel field from a single fixed-size model.

Figure 3 further illustrates this process, showing the evolution of the field during the ODE integration. The periodic shift applied at each step is visualized, demonstrating how local information is communicated globally, ensuring phase coherence across all scales.

Figure 5 provides a detailed look at the physical statistics of our generated fields. The power spectrum $P(k)$ and cross-correlation $C(k)$ show excellent agreement with the ground truth across all relevant scales. The transfer function $T(k)$ is close to unity, indicating that the model is not systematically biased in its reproduction of structure at different scales.

Generation Efficiency

Table 3 compares generation time and memory across resolutions. The speedup over diffusion arises from two independent factors: rectified flow integration converges in 25–100 steps versus hundreds for DDPM, and each step processes $8\times$ spatially compressed wavelet patches (64^3) rather than pixel-space sub-volumes (128^3). Crucially, the patch-based architecture maintains a constant memory footprint at all resolutions, enabling generation at 1024^3 where pixel-space diffusion exhausts available GPU memory.

6 Conclusion

We presented **Cosmo3DFlow-Scale**, a generative framework that directly resolves two fundamental barriers that have prevented field reconstruction at gigavoxel scale: GPU memory saturation and inter-patch spatial incoherence. Lossless wavelet compression bounds training memory to the patch size independently of target resolution, and Wavelet-Flow

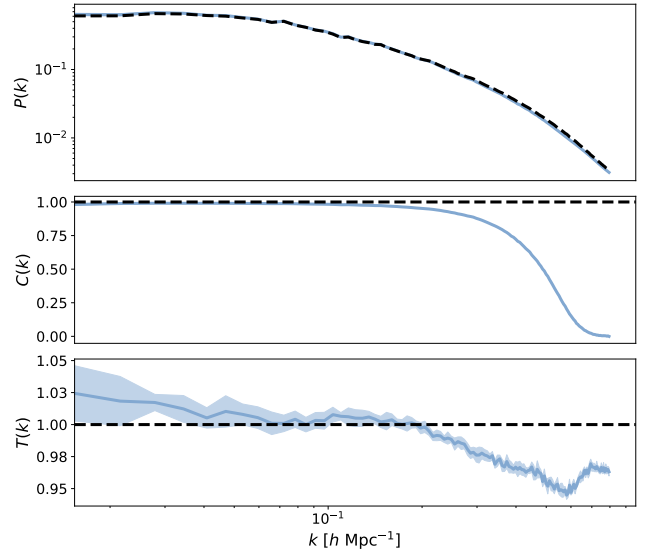


Figure 5: Physics validation metrics at 256^3 . Top row: Power spectrum $P(k)$ and Cross-correlation $C(k) \approx 1$ across scales. Bottom row: Transfer function $T(k)$ and a 2D joint probability distribution of voxel values, showing excellent agreement with the target distribution across multiple orders of magnitude.

Table 3: Sampling efficiency and memory footprint (A100-80GB). Memory (M) in GB, Time (T) in seconds.

Resolution	Batch	CosmoFOLD		Ours	
		T(s)	M(G)	T(s)	M(G)
256^3	4	112	31.4	25	10.8
	8	101	61.7	23.8	10.8
	16	101	61.7	22	10.8
512^3	4	893	33.1	185	8.1
	8	770	66.3	169	13.3
	16	Out of	Memory	163	22.8
1024^3	1	12141	24.4	2323	34.8
	4	7078	47.1	1482	38.8
	8	Out of	Memory	1351	38.8
	16	Out of	Memory	1302	43.8

Tiling with shift-compensated positional encoding maintains global phase coherence during deterministic ODE integration. The result is a single model that generates coherent 1024^3 dark matter density fields with a $\sim 5\times$ reduction in generation time over diffusion-based tiling, and that scales further via distributed patch-parallel execution.

The broader significance lies in enabling statistically rigorous ensemble inference at resolutions demanded by DESI, Euclid, and LSST. Generating 10–50 posterior samples at 1024^3 on a single GPU would require days; with distributed execution across P GPUs the inner loop scales as $1/P$, making posterior uncertainty quantification computationally tractable at gigavoxel scale for the first time.

Limitations and Ethical Considerations

The model is trained at a single fiducial cosmology; cosmology-conditioned training across the Quijote LH is left for future work. The synthetic simulation data poses no privacy or ethical concerns. Code and model weights will be released for reproducibility.

GenAI Disclosure

Google Gemini was used for initial figure drafting. Claude was used for text polishing.

Acknowledgments

This work is partly supported by NSF-Simons AI Institute for Cosmic Origins (CosmicAI: Grant 2421782).

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A Appendix

Table 4 summarizes the notation used throughout this paper.

Table 4: Notation and key definitions.

Symbol	Definition
<i>Problem Setup</i>	
$\mathbf{x} \in \mathbb{R}^{N^3}$	Initial conditions (latent, $z = 127$)
$\mathbf{y} \in \mathbb{R}^{N^3}$	Observations (evolved, $z = 0$)
N	Grid resolution per dimension
p	Patch size (default 128)
$p(\mathbf{x} \mathbf{y})$	Posterior of initial conditions
$\delta = \rho/\bar{\rho} - 1$	Density contrast field
<i>Flow Matching</i>	
$t \in [0, 1]$	Flow time
$\mathbf{x}_0 \sim p_0$	Prior sample (Gaussian noise)
$\mathbf{x}_1 \sim p_1$	Target sample
$\mathbf{x}_t$	Interpolant: $t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$
$\mathbf{u}_t$	Target velocity: $\mathbf{x}_1 - \mathbf{x}_0$
$\mathbf{v}_\theta$	Learned velocity network
K	ODE integration steps
θ	Cosmology parameters $[\Omega_m, \Omega_b, h, n_s, \sigma_8]$
<i>Wavelet Transform (Haar)</i>	
DWT / IDWT	Discrete Wavelet Transform / Inverse
$\tilde{\mathbf{x}}, \tilde{\mathbf{y}}$	Wavelet-transformed fields
$\mathbf{c}_{LLL}$	Approximation coefficients
$\mathbf{c}_{LLH}, \dots, \mathbf{c}_{HHH}$	7 detail coefficient bands
<i>Tiling and Positional Encoding</i>	
$\boldsymbol{\eta}^k$	Full-volume field at flow step k
$\boldsymbol{\xi}$	Full-volume conditioning field
$\mathbf{o}_j$	Unshifted patch origin (d_0, h_0, w_0)
$\mathbf{pe}$	Sinusoidal PE tensor $(1, 3, p, p, p)$
n	Number of PE frequency bands
$f_j = \tau_{\max}^{-j/(n-1)}$	PE frequencies
<i>Evaluation Metrics</i>	
VRMSE	Variance-normalized RMSE
$P(k)$	Isotropic power spectrum
$C(k)$	Cross-correlation coefficient
$T(k)$	Transfer function